THE NOVA RESEARCH CURRICULUM

Biology

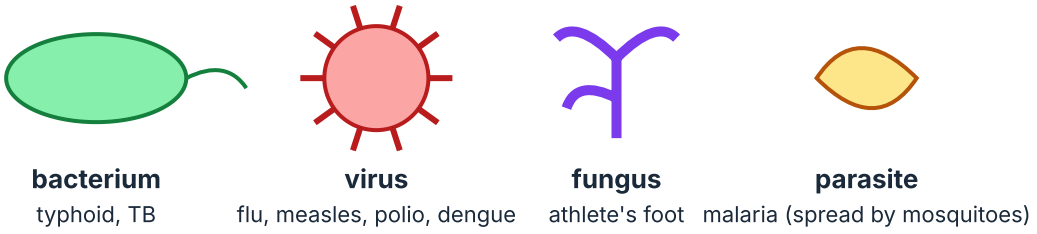
Grade 7 · Term 3

Student edition

Outcome: BI-7.3 immunity, pathogens, infectious diseases and vaccines.

Lesson notes

- **Pathogens (Weeks 1–4):** microbes that cause disease: bacteria (typhoid, tuberculosis), viruses (flu, measles, polio, dengue), fungi (athlete's foot), and the malaria parasite. Spread by air, water, food, touch and insects (mosquitoes for malaria and dengue). Prevention: hand-washing, clean water, covering coughs, removing standing water.



Four kinds of pathogen (not to scale; viruses are far smaller than bacteria).

- **Immunity (Weeks 5–8):** first defences: skin, mucus, stomach acid. White blood cells engulf germs and make antibodies that fit a particular germ. Memory cells remember the germ, so you fight it faster next time.

  ▶ **Try it on NovaSchool:** White blood cells at work — open **Simulations** on your dashboard.

- **Vaccines (Weeks 9–12):** a vaccine contains a harmless form or part of a germ, so the body makes antibodies and memory cells without getting ill. Pakistan's routine childhood vaccinations and polio campaigns. Antibiotics kill bacteria but do not work on viruses; finish the course the doctor gives.

Workbook pages

Page	Task
1	Pathogen table: type, disease, spread, prevention.
2	Hand-washing glitter test record.
3	How a vaccine works: four-step comic strip.

Question bank

#	Outcome	Question	Your answer
1	BI-7.3	Dengue is spread by ...	dirty water / mosquitoes / touch / food
2	BI-7.3	Polio is caused by a ...	bacterium / fungus / virus / parasite
3	BI-7.3	Which cells make antibodies?	red blood cells / platelets / white blood cells / nerve cells
4	BI-7.3	Antibiotics work against viruses. True or false.	True / False
5	BI-7.3	A vaccine contains ...	a strong live germ / a harmless form or part of a germ / antibiotics / vitamins only
6	BI-7.3	Skin is one of the body's first defences. True or false.	True / False
7	BI-7.3	Give two ways to stop dengue spreading.	_____
8	BI-7.3	Why are you less likely to catch measles twice?	_____